

Date: 23/07/2025

Lab Practical #8:

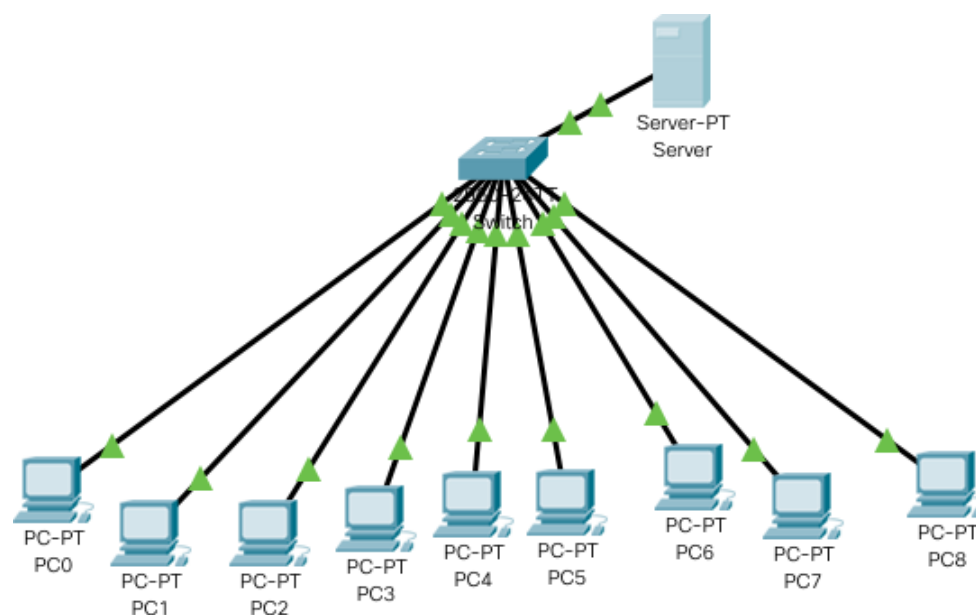
Study & Survey of Institute organization network infrastructure.

Practical Assignment #8:

1. Identify type of network in your institute. Draw a design of network in your institute (Any Lab/Floor/Building).

Lab No: C- 302

Topology: Star Topology



2. List how many network devices and types of cable used and give its details.

Network Devices:

1. **Router**
 - Connects different networks and routes data packets between them, such as between a local network and the internet.
2. **Switch**
 - Connects multiple devices within the same network and directs data to the correct destination device.
3. **Firewall**
 - Provides network security by monitoring and controlling incoming and outgoing traffic based on predetermined rules.
4. **Access Point (AP)**
 - Provides wireless network connectivity for devices to connect to a wired network.
5. **Network Interface Card (NIC)**
 - Hardware installed in devices to enable network connectivity.
6. **Modem**



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- Modulates and demodulates signals for internet access over telephone lines or other communication mediums.
- 7. Load Balancer**
 - Distributes network or application traffic across multiple servers to ensure reliability and performance.

Types of Cables:

- 1. Ethernet Cable (Twisted Pair - Cat5e, Cat6, Cat6a)**
 - Used for wired network connections; twisted pairs reduce interference and support high-speed data transfer.
- 2. Fiber Optic Cable**
 - Uses light to transmit data at very high speeds over long distances with minimal signal loss.
- 3. Coaxial Cable**
 - Used in some legacy network setups; carries data via electrical signals within a single conductor surrounded by shielding.
- 4. Console Cable (RJ45 to Serial)**
 - Used to connect network devices like switches and routers to a computer for configuration.